

This practice is built exactly like the real test — same questions, same order, different numbers. Work every problem, then check the worked key. Anything you miss is exactly what to study.

1. Simplify: $\sqrt{80}$

2. Simplify: $\sqrt{108}$

3. Simplify: $\sqrt{(64k^2)}$

4. Simplify: $\sqrt{6} \cdot \sqrt{3}$

5. Simplify: $5\sqrt{7} + 2\sqrt{7} - 3\sqrt{7}$

6. Simplify: $\sqrt{50} + 4\sqrt{2}$

7. Rationalize the denominator: $12/\sqrt{6}$

8. Rationalize the denominator: $4/\sqrt{20}$

Algebra 1 – Module 8 Practice Test

Full Name: _____

Period: _____

9. Simplify: $(5n^3)^3$

10. Simplify. Write your answer with a positive exponent: n^2 / n^8

11. Error Analysis — Rex evaluated $\sqrt{64 + 225}$ as $8 + 15 = 23$. Identify his mistake, then evaluate correctly.

12. Evaluate each expression.

a. $81^{(1/2)}$

b. $216^{(1/3)}$

13. Solve: $\sqrt{x - 7} = 8$

Cumulative Review — ACT Style. Circle the best answer.

14. Which ordered pair solves the system $x + y = 10$ and $x - y = 4$?

- (A) (6, 4)
- (B) (7, 3)
- (C) (3, 7)
- (D) (8, 2)

15. Solve the compound inequality: $3 < x + 4 \leq 9$

- (E) $7 < x \leq 13$
- (F) $-1 \leq x < 5$
- (G) $-1 < x \leq 5$
- (H) $3 < x \leq 9$

16. Which line is parallel to $y = -2x + 5$ through (0, -4)?

- (A) $y = 2x - 4$
- (B) $y = -(1/2)x - 4$
- (C) $y = -2x - 4$
- (D) $y = -2x + 4$